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Higher-order Weierstrass weights of branch points on superelliptic curves

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Abstract. In this paper we consider the problem of calculating the higher-order Weierstrass weight of the branch points of a superelliptic curve C. For any $q > 1$, we give an exact formula for the q-weight of an affine branch point. We also find a formula for the q-weight of a point at infinity in the case where n and d are relatively prime. With these formulas, for any fixed n, we obtain an asymptotic formula for the ratio of the q-weight of the branch points, denoted BW_q, to the total q-weight of points on the curve:

$$\liminf_{d\to\infty} \frac{BW_q}{g(g-1)^2(2q-1)^2} \geq \frac{n+1}{3(n-1)^2(2q-1)^2},$$

with equality when the limit is taken such that $\gcd(n,d) = 1$.

1. Introduction

Let $q \in \mathbb{N}$. A q-Weierstrass point (or higher-order Weierstrass point) is a point P on a curve for which there exist holomorphic q-differentials that have higher than expected orders of vanishing at P. Each q-Weierstrass point has an associated q-weight, denoted $w^{(q)}(P)$, which measures how much higher than expected those orders of vanishing are. A curve of genus $g \geq 2$ has finitely many q-Weierstrass points.

Importantly, the q-weight of a point is invariant under automorphism. Thus, higher-order Weierstrass points are important in the study of automorphisms of algebraic curves. For instance, Lewittes showed in [**6**] that if an automorphism has at least five fixed points, then all of its fixed points are 1-Weierstrass points. Further, Mumford, in [**8**], has suggested that q-Weierstrass points on an algebraic curve are analogous to q-torsion points on an elliptic curve. For more on the history of Weierstrass points, we refer the reader to [**2**]. For background material of Weierstrass points specifically on superelliptic curves, see [**12**].

Let C be a curve of genus $g \geq 2$ of the form $y^n = f(x)$ where $f(x)$ is a separable polynomial of degree $d > n \geq 2$. Such a curve is said to be superelliptic. In the cover $C \to \mathbb{P}^1$, the points above the roots of $f(x)$ are branch points. If $n \nmid d$, the point (or points) above ∞ in the nonsingular model of C is also a branch point. One can show that each branch point is a q-Weierstrass point for all q; in the case

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where $n = 2$, the branch points are exactly the 1-Weierstrass points. Let B be an affine branch point on C and, if $n \nmid d$, P^∞ a nonsingular branch point at infinity. In [**16**, Theorem 8], Towse calculated the 1-weight of the branch points (affine and at infinity) as a function of n and d. In the case of $\gcd(n, d) = 1$, he found that

$$w^{(1)}(B) = \frac{g(n+1)(d-7)}{12} + (d-1)\sum_{j=1}^{n-1}\left\{\frac{-dj}{n}\right\} j,$$

where $\{x\}$ denotes the fractional part of x, and

$$w^{(1)}(P^\infty) = \frac{(n^2-1)(d^2-1)}{24} - g.$$

Given that the total 1-weight of points on a curve of genus g is $g^3 - g$, he was able to calculate the fraction that the branch points' 1-weight (denoted BW) accounted for as

$$\lim_{d\to\infty} \frac{BW}{g^3 - g} \geq \frac{n+1}{3(n-1)^2},$$

with equality when the limit is taken over integers d such that $\gcd(n, d) = 1$.

The goal of this paper is to extend Towse's results to higher-order Weierstrass weights of branch points on a superelliptic curve. To achieve this, we first produce a basis for the space of holomorphic q-differentials on a superelliptic curve C. A common approach to calculate the q-weight is to work with the Wronskian of this basis, a method first described by Hurwitz in [**5**]. However, we take a different approach, instead using results from numerical semigroups and non-representable numbers. In this way we obtain a formula for the q-weight of an affine branch point as a function of n, d, and q. The main result is Theorem 2. In particular, when $\gcd(n, d) = 1$ we find for $q \geq 2$

$$w^{(q)}(B) = \frac{g(n+1)(d-7)}{12} + g + (d-1)\sum_{j=1}^{n-1}\left\{-\frac{(d+1)q+dj}{n}\right\} j.$$

As for the points at infinity, with two examples, we show that if $\gcd(n, d) > 1$, one cannot get a formula for $w^{(q)}(P^\infty)$ based only on n, d, and q. However, if $\gcd(n, d) = 1$ and $q \geq 2$, then in Theorem 3 we have

$$w^{(q)}(P^\infty) = \frac{(n^2-1)(d^2-1)}{24}.$$

Since the total q-weight of points on a curve of genus g is $g(g-1)^2(2q-1)^2$ for $q \geq 2$, we get an similar asymptotic result in Proposition 6 for the proportion of branch points' q-weight (denoted BW_q). We find

$$\lim_{d\to\infty} \frac{BW_q}{g(g-1)^2(2q-1)^2} \geq \frac{n+1}{3(n-1)^2(2q-1)^2},$$

with equality when the limit is taken over integers d such that $\gcd(n, d) = 1$.

The paper is organized as follows. In Section 2, we provide some background material on calculating the q-Weierstrass weight of a point. We also include some notation and results for non-representable integers in numerical semigroups with two generators. In Section 3, we find a basis for the space of holomorphic q-differentials on a curve C by presenting a set of linearly independent holomorphic differentials and then counting them to make sure there are as many as the Riemann-Roch theorem predicts. In Section 4, we have our main results. We find a

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formula for the q-weight of an affine branch point, and we use that to derive a few corollaries for specific cases of n and d. We also note that, for given n and d, the q-weight of a branch point depends only on the value of q modulo n. We also give examples to show that if $\gcd(n,d) > 1$ the q-weight of a point at infinity cannot necessarily be determined just by knowing n and d. If $\gcd(n,d) = 1$, however, we can calculate the weight. In both cases, we obtain some asymptotic results about the proportion of q-weight that the branch points contain.

2. Preliminaries and notation

2.1. q-Weierstrass points. In this paper, we will follow the approach given in [**13**, Section 2]. We describe the notation and major results for calculating weights of q-Weierstrass points here.

Let k be an algebraically closed field, C be a non-singular projective curve over k of genus $g \geq 2$, and $k(C)$ its function field. For any $f \in k(C)$, let $\operatorname{div}(f)$ denote the divisor associated to f. For any divisor $D = \sum_P n_P P$ and any point P, let $\nu_P(D) = n_P$, and let $\operatorname{ord}_P(f) = \nu_P(\operatorname{div}(f))$.

For any $q \in \mathbb{N}$, let $H^0(C, (\Omega^1)^q)$ be the $\mathbb{C}$-vector space of holomorphic q-differentials on C, a space of dimension

$$d_q = \begin{cases} g & \text{if } q = 1, \\ (g-1)(2q-1) & \text{if } q \geq 2. \end{cases}$$

For P a degree 1 point on C, consider a basis $\{\psi_1, \ldots, \psi_{d_q}\}$ of $H^0(C, (\Omega^1)^q)$ where

$$\operatorname{ord}_P(\psi_1) < \operatorname{ord}_P(\psi_2) < \cdots < \operatorname{ord}_P(\psi_{d_q}).$$

The q-weight of P is

$$w^{(q)}(P) = \sum_{i=1}^{d_q} \operatorname{ord}_P(\psi_i) - \sum_{j=0}^{d_q-1} j.$$

We call the point P a *q-Weierstrass point* if $w^{(q)}(P) > 0$.

PROPOSITION 1. [**4**, III.5.10] *Let C be a curve of genus $g \geq 2$ and let $q \geq 1$. Then the total q-weight of points on C is*

$$\sum_{P \in C} w^{(q)}(P) = (g-1)d_q(2q-1+d_q) = \begin{cases} g^3 - g & \text{if } q = 1, \\ g(g-1)^2(2q-1)^2 & \text{if } q \geq 2. \end{cases}$$

2.2. Non-representable numbers. For notation, let $\mathbb{N}_0$ be the set of non-negative integers. Let $a, b \in \mathbb{N}$ and consider the set

$$R(a,b) = \{ax + by : x, y \in \mathbb{N}_0\}.$$

Elements of $R(a,b)$ are called *(a,b)-representable numbers*. The complement of $R(a,b)$ in $\mathbb{N}_0$, denoted $NR(a,b)$, is the set of *(a,b)-representable numbers*. When there is no confusion, we will omit the (a,b) and simply refer to these numbers as representable or non-representable.

The problem of calculating the cardinality of $NR(a,b)$ dates to the late 19th century in [**15**]. Clearly, if $\gcd(a,b) > 1$ then $NR(a,b)$ is an infinite set. It is straightforward to show that the converse is true too. For example, see [**10**, Theorem 1.0.1] for two proofs of the following result.

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LEMMA 1. *For $a, b \in \mathbb{N}$, if $\gcd(a,b) = 1$ then $NR(a,b)$ is a finite set.*

For the rest of this section, we will assume $\gcd(a,b) = 1$. Thus, $NR(a,b)$ is finite and so we can compute its cardinality.

PROPOSITION 2. [**15**, Page 134] *For $a, b \in \mathbb{N}$ with $\gcd(a,b) = 1$,*

$$|NR(a,b)| = \frac{(a-1)(b-1)}{2}.$$

This result is important in the theory of algebraic curves. Suppose a plane curve C is given by the affine equation

$$\alpha_{a,0}x^a + \alpha_{0,b}y^b + \sum_{i,j} \alpha_{i,j}x^i y^j = 0$$

for constants $\alpha_{i,j}$ with $\alpha_{a,0} \cdot \alpha_{0,b} \neq 0$ and where the summation is over non-negative i, j such that $aj + bi < ab$. Such a curve is called a $C_{a,b}$ curve. These curves can be seen as a generalization of elliptic and hyperelliptic curves in Weierstrass form. With the Riemann-Roch Theorem, if the affine part of the curve is non-singular then one can show that the genus of such a curve is exactly $(a-1)(b-1)/2$, the cardinality of $NR(a,b)$. For details, see [**7**] or [**14**].

For the purposes of this paper, we will also need to know the sum of the elements of $NR(a,b)$. This problem was solved in [**1**] using generating functions.

PROPOSITION 3. *For $a, b \in \mathbb{N}$ with $\gcd(a,b) = 1$,*

$$\sum_{n \in NR(a,b)} n = \frac{(a-1)(b-1)(2ab-a-b-1)}{12}.$$

This result was generalized to a formula for the sum of the mth powers of elements of $NR(a,b)$. See [**11**] or [**17**] for details.

3. A basis of holomorphic q-differentials

In this section, we give a basis for the space of holomorphic q-differentials on a superelliptic curve C. The main result of this section is as follows:

THEOREM. *Let C be a curve of genus $g \geq 2$ given in affine coordinates by $y^n = f(x)$, for $f(x)$ a separable polynomial of degree $d > n \geq 2$. For $q \geq 1$, let $H^0(C, (\Omega^1)^q)$ be the space of holomorphic q-differentials on C. Let*

$$\mathfrak{B}_{n,d,q} = \left\{ x^i y^j \left(\frac{\mathrm{d}x}{y^{n-1}} \right)^q : 0 \leq i,\ 0 \leq j < n,\ ni + dj \leq (2g-2)q \right\}.$$

Then $\mathfrak{B}_{n,d,q}$ is a basis for $H^0(C, (\Omega^1)^q)$.

In order to prove this, we need the following results (based on the work in [**16**]) and a useful lemma.

Let $G = \gcd(n,d)$. For g the genus of C, we have $2g - 2 = nd - n - d - G$. Let $\{\alpha_1, \ldots, \alpha_d\}$ denote the d distinct roots of $f(x)$ and let $B_i = (\alpha_i, 0)$ for $i = 1, \ldots, d$. For each non-root ω of $f(x)$, let $P_1^\omega, \ldots, P_n^\omega$ denote the n points on C over $x = \omega$. And let $P_1^\infty, \ldots, P_G^\infty$ denote the G points over ∞ in the non-singular model of C. One then has the following principal divisors.

- $\operatorname{div}(y) = \sum_{j=1}^{d} B_j - \frac{d}{G} \sum_{m=1}^{G} P_m^\infty,$

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- $\operatorname{div}(x-\alpha_i) = nB_i - \dfrac{n}{G}\displaystyle\sum_{m=1}^{G} P_m^{\infty}$,
- $\operatorname{div}(x-\omega) = \displaystyle\sum_{j=1}^{n} P_j^{\omega} - \dfrac{n}{G}\sum_{m=1}^{G} P_m^{\infty}$.
- $\operatorname{div}(\mathrm{d}x) = (n-1)\displaystyle\sum_{j=1}^{d} B_j - \left(\dfrac{n}{G}+1\right)\sum_{m=1}^{G} P_m^{\infty}$,

From these, we see that $\operatorname{div}((\mathrm{d}x/y^{n-1})^q) = \frac{(2g-2)q}{G}\sum_{m=1}^{G} P_m^{\infty}$.

For integers i, j, let $f_{i,j} = x^i y^j (\mathrm{d}x/y^{n-1})^q$. We want to find conditions on i and j such that $f_{i,j}$ is a holomorphic q-differential. Note that $f_{i,j}$ can have poles only at the points above ∞ if $i, j \geq 0$. In that situation, we find

$$\operatorname{ord}_{P_m^{\infty}}(f_{i,j}) = \frac{(2g-2)q-(ni+dj)}{G}$$

for each pair (i, j) and each m. Hence $f_{i,j}$ is a holomorphic q-differential as long as $i \geq 0, j \geq 0$, and $ni + dj \leq (2g-2)q$.

LEMMA 2. *Let $n, d, q \in \mathbb{Z}$ with $2 \leq n < d$ and $q \geq 2$. As above, let $G = \gcd(n, d)$ and $2g-2 = nd-n-d-G$. For all but finitely many triples (n, d, q), one has*

$$(2g-2)q - d(n-1) \geq 0.$$

The exceptional cases are $(n, d, q) \in \{(2, 5, 2), (2, 6, 2)\}$.

PROOF. First, note that $(2g-2)q - d(n-1) = (nd-n-d-G)q - d(n-1) = d(n-1)(q-1) - q(n+G)$. Thus, to show our desired inequality, it is equivalent to show

$$d \geq \left(\frac{q}{q-1}\right)\left(\frac{n+G}{n-1}\right).$$

For notation, let $h(q, n) = \left(\frac{q}{q-1}\right)\left(\frac{n+G}{n-1}\right)$. We aim to show $d \geq h(q, n)$.

For $q \geq 2$, the maximum value of $q/(q-1)$, which occurs when $q = 2$, is 2. For $n \geq 2$, the maximum value of $(n+G)/(n-1)$ occurs when G is largest, so when $G = n$, and when $n = 2$. The maximum value is 4. Thus $h(q, n) \leq 2 \cdot 4 = 8$ so $d \geq h(q, n)$ for all $d \geq 8$.

Now we consider cases of n. If $n \geq 4$, then $G = n$ or $G < n$. If $G = n$, then $n|d$, so $d \geq 2n \geq 8$, so $d \geq h(q, n)$. If $G < n$, then $G \leq n/2$, so $h(q, n) \leq 2(n+n/2)/(n-1) = 3 + 3/(n-1) \leq 4$. Since $d > n$, we have $d > n \geq 4 \geq h(q, n)$, as desired.

If $n = 3$, then $G = 1$ or $G = 3$. If $G = 1$, then $h(q, 3) \leq 2 \cdot 2 = 4$. In this case, since $d > n = 3$, we have $d \geq 4$, so $d \geq h(q, 3)$. If $G = 3$, then $h(q, 3) \leq 2 \cdot 3 = 6$. In this case, $d \geq 2n = 6 \geq h(q, 3)$, as desired.

If $n = 2$, then $G = 1$ or $G = 2$. Note that to have $g \geq 2$, we only consider $d \geq 5$. If $G = 1$ and $q = 2$, then $h(2, 2) = 2 \cdot 3 = 6$. Since $G = 1$, d is odd, so $d \geq h(2, 2)$ for all d except for $d = 5$. If $G = 1$ and $q \geq 3$, then $h(q, 2) \leq (3/2) \cdot 3 = 9/2$, so $d \geq h(q, 2)$ for all $d \geq 5$. If $G = 2$ and $q = 2$, then $h(2, 2) = 2 \cdot 4 = 8$. Since $G = 2$, d is even, so $d \geq h(2, 2)$ for all d except for $d = 6$. If $G = 2$ and $q \geq 3$, then $h(q, 2) \leq (3/2) \cdot 4$, so $d \geq h(q, 2)$ for all $d \geq 6$.

Thus, the only exceptional cases are $(n, d, q) = (2, 5, 2)$ or $(2, 6, 2)$. For all other triples, we find that $d \geq h(q, n)$, or, equivalently, that

$$(2g-2)q - d(n-1) \geq 0,$$

as desired. □

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THEOREM 1. *Let C be a curve of genus $g \geq 2$ given in affine coordinates by $y^n = f(x)$, for $f(x)$ a separable polynomial of degree $d > n$. For $q \geq 1$, let $H^0(C, (\Omega^1)^q)$ be the space of holomorphic q-differentials on C. Let*

$$\mathfrak{B}_{n,d,q} = \left\{ x^i y^j \left(\frac{\mathrm{d}x}{y^{n-1}} \right)^q : 0 \leq i,\, 0 \leq j < n,\, ni + dj \leq (2g-2)q \right\}.$$

Then $\mathfrak{B}_{n,d,q}$ is a basis for $H^0(C, (\Omega^1)^q)$.

PROOF. The $q = 1$ case, in a slightly different form, is proved in [**16**]. For completeness, we will first prove the $q \geq 2$ case here and then adapt our argument to cover the $q = 1$ case.

Suppose $q \geq 2$. With the restriction that $0 \leq j < n$, we see that these holomorphic q-differentials are linearly independent. We therefore need to show that $|\mathfrak{B}_{n,d,q}| = d_q = (2q-1)(g-1)$.

We first consider the case where $(n, d, q) \notin \{(2, 5, 2), (2, 6, 2)\}$ and let $\mathfrak{B} = \mathfrak{B}_{n,d,q}$. Note that we require $i \geq 0$ and $ni + dj \leq (2g-2)q$, so

$$0 \leq i \leq \left\lfloor \frac{(2g-2)q - dj}{n} \right\rfloor.$$

For each $j = 0, \ldots, n-1$, we have $(2g-2)q - dj \geq (2g-2)q - d(n-1)$, which is non-negative by Lemma 2. (This is why we handle the $(2, 5, 2)$ and $(2, 6, 2)$ cases separately.) Thus, to calculate the number of pairs (i, j) in $\mathfrak{B}$, we will let j go from 0 to $n-1$ and count the number of indices i that correspond to each j value. I.e.,

$$|\mathfrak{B}| = \sum_{j=0}^{n-1} \left(1 + \left\lfloor \frac{(2g-2)q - dj}{n} \right\rfloor \right).$$

Since $\lfloor x \rfloor = x - \{x\}$, we simplify the sum to get

$$|\mathfrak{B}| = n + (2g-2)q - \frac{d(n-1)}{2} - \sum_{j=0}^{n-1} \left\{ \frac{(-d-G)q - dj}{n} \right\}.$$

Now, we consider cases of G. If $G = n$, then $n|d$, so $n|((-d-G)q - dj)$, so each term in the summation is 0. Note that $n - d(n-1)/2 = -(nd - d - 2n)/2 = -(nd - n - d - G)/2 = -(g-1)$. Then $|\mathfrak{B}| = (2g-2)q - (g-1) = (2q-1)(g-1) = d_q$, as desired.

Next, suppose $G \neq n$. Let $n' = n/G$ and $d' = d/G$. Dividing the numerator and denominator by G, the summation equals $\sum_{j=0}^{n-1} \left\{ \frac{(-d'-1)q - d'j}{n'} \right\}$. Since $\gcd(n', d') = 1$, as j goes from 0 to $n'-1$ modulo n', the numerators are distinct modulo n' and therefore in every congruence class exactly once modulo n'. Since $n/n' = G$, this summation equals $G \sum_{k=0}^{n'-1} \frac{k}{n'} = G(n'-1)/2$. All together,

$$|\mathfrak{B}| = n + (2g-2)q - d(n-1)/2 - (n-G)/2.$$

I.e. $|\mathfrak{B}| = n + 2q(g-1) - (1/2)(nd - d + n - G) = 2q(g-1) - (1/2)(nd - d - n - G) = (2q-1)(g-1) = d_q$, as desired.

To complete the proof for $q \geq 2$, we consider the exceptional cases. Suppose $(n, d, q) = (2, 5, 2)$, so $g = 2$ and $d_2 = 3$. Then

$$\mathfrak{B}_{2,5,2} = \left\{ x^i y^j (\mathrm{d}x/y)^2 : i \geq 0, 0 \leq j < 2, 2i + 5j \leq 4. \right\}$$

Thus, $\mathfrak{B}_{2,5,2} = \{(\mathrm{d}x/y)^2,\, x(\mathrm{d}x/y)^2,\, x^2(\mathrm{d}x/y)^2\}$, so $|\mathfrak{B}_{2,5,2}| = 3 = d_2$, as desired.

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Suppose $(n, d, q) = (2, 6, 2)$, so $g = 2$ and $d_2 = 3$. Then

$$\mathfrak{B}_{2,6,2} = \left\{x^i y^j (\mathrm{d}x/y)^2 : i \geq 0, 0 \leq j < 2, 2i + 6j \leq 4.\right\}$$

Thus, $\mathfrak{B}_{2,6,2} = \{(\mathrm{d}x/y)^2,\ x(\mathrm{d}x/y)^2,\ x^2(\mathrm{d}x/y)^2\}$, so $|\mathfrak{B}_{2,6,2}| = 3 = d_2$, as desired.

Now, suppose $q = 1$. Following the approach above, given j we need integers i such that

$$0 \leq i \leq \left\lfloor \frac{(2g-2) - dj}{n} \right\rfloor.$$

If $j \leq n-2$, then $(2g-2) - dj \geq (2g-2) - d(n-2) = d - n - G \geq 0$ since $d \geq n$. If $j = n-1$, then $(2g-2) - d(n-1) = -n - G < 0$, so there are no such i. Thus, our summation for $|\mathfrak{B}|$ ends at $j = n-2$ instead of $j = n-1$. Since we have a formula for the summation above, we can subtract the $j = n-1$ term out front to get

$$|\mathfrak{B}| = -\left(1 + \left\lfloor \frac{-n-G}{n} \right\rfloor\right) + \sum_{j=0}^{n-1} \left(1 + \left\lfloor \frac{(2g-2) - dj}{n} \right\rfloor\right).$$

Since $q = 1$ the summation equals $g - 1$, so $|\mathfrak{B}| = -(1-2) + (g-1) = g = d_1$, as desired. □

4. Weights of branch points

In this section, we use the bases we found in the previous section to calculate the q-weight of the affine branch points and, in the case that $\gcd(n, d) = 1$, the point at infinity.

4.1. Weights of affine branch points. Suppose $q \geq 2$. For C given by $y^n = f(x)$ with $f(x)$ separable of degree d, let α be a root of $f(x)$. Then $B = (\alpha, 0)$ is an affine branch point of C. Note that we can replace x by $(x - \alpha)$ in our basis $\mathfrak{B}_{n,d,q}$ to produce a new basis $\mathfrak{B}_{n,d,q,\alpha}$. That is,

$$\mathfrak{B}_{n,d,q,\alpha} = \{(x-\alpha)^i y^j (\mathrm{d}x/y^{n-1})^q : i \geq 0,\ 0 \leq j < n,\ ni + dj \leq (2g-2)q\}$$

is a basis for $H^0(C, (\Omega^1)^q)$.

Let $f_{i,j,\alpha} = (x-\alpha)^i y^j (\mathrm{d}x/y^{n-1})^q \in \mathfrak{B}_{n,d,q,\alpha}$. Then

$$\nu_B(f_{i,j,\alpha}) = ni + j.$$

Since $0 \leq j < n$, these valuations are all different, and thus

$$w^{(q)}(B) = \sum_{(i,j)\in S} (ni + j) - \sum_{k=0}^{d_q - 1} k,$$

where $S = \{(i, j) \in \mathbb{Z}^2 : i \geq 0,\ 0 \leq j < n,\ ni + dj \leq (2g-2)q\}$. We rewrite this as $w^{(q)}(B) = W_1 - W_2 - W_3$ where

$$W_1 = \sum_{(i,j)\in S} (ni + dj), \quad W_2 = (d-1) \sum_{(i,j)\in S} j, \quad W_3 = \sum_{k=0}^{d_q - 1} k. \tag{1}$$

We have

$$W_3 = \frac{(d_q - 1)(d_q)}{2} = \frac{1}{2}\left((2g-2)^2 q^2 + (2g-2)(1-2g)q + g(g-1)\right).$$

We will evaluate W_1 and W_2 with the following propositions.

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PROPOSITION 4. *Let $n, d, q \in \mathbb{N}$ such that $n < d$ and $q \geq 2$. Then*

$$W_1 = 2(g-1)^2q^2 + (g-1)Gq + \frac{G^2 - 1 - (n-1)(d-1)(2nd - n - d - 1)}{12}.$$

We will first sketch the proof in the situation where $\gcd(n, d) = 1$. Afterward, we will prove the theorem for any gcd.

When $\gcd(n, d) = G = 1$, for $(i, j) \in S$, the terms $ni + dj$ are distinct integers from 0 to $(2g-2)q$. From Proposition 2, since $(2g-2)q \geq nd - n - d$ (by Lemma 3 below), all of the $(n-1)(d-1)/2$ (n, d)-non-representable integers are in that interval. The sum of the non-representable integers, as is given in Proposition 3, is $(n-1)(d-1)(2nd - n - d - 1)/12$. Thus, if $\gcd(n, d) = 1$, we add up all of the integers from 0 to $(2g-2)q$ and subtract off the non-representable integers to get $W_1 = (2g-2)q((2g-2)q + 1)/2 - (n-1)(d-1)(2nd - n - d - 1)/12$.

If $\gcd(n, d) > 1$, then the terms $ni + dj$ are no longer distinct, so we need to evaluate the sum more carefully.

PROOF. Let $G = \gcd(n, d)$. First, we observe that $W_1 = G\sum_{(i,j)\in S}(n'i + d'j)$ for $n' = n/G$ and $d' = d/G$. Note that $\gcd(n', d') = 1$. For k from 0 to $G - 1$, let

$$S_k = \{(i, j) : i \geq 0,\ kn' \leq j < (k+1)n',\ ni + dj \leq (2g-2)q\}.$$

In particular, S is the disjoint union of the sets S_k. Let $W_{1,k} = \sum_{(i,j)\in S_k}(n'i + d'j)$. Then

$$W_1 = G\sum_{k=0}^{G-1} W_{1,k}.$$

Letting $j' = j - kn'$, we rewrite S_k as

$$S_k = \{(i, j' + kn') : i \geq 0,\ 0 \leq j' < n',\ ni + dj' \leq (2g-2)q - n'dk\}$$

and dividing the last inequality through by G we obtain

$$S_k = \{(i, j' + kn') : i \geq 0,\ 0 \leq j' < n',\ n'i + d'j' \leq \frac{(2g-2)}{G}q - n'd'k\}.$$

Let $m_k = \frac{(2g-2)}{G}q - n'd'k$, the upper bound in S_k. The following lemma will allow us to conclude that all of the (n', d')-non-representable integers are less than m_k.

LEMMA 3. *Let $m_k = \frac{(2g-2)}{G}q - n'd'k$. Then $m_k \geq n'd' - n' - d'$ for all $n, d, q \in \mathbb{N}$ with $0 \leq k \leq G - 1$, $n < d$, $g \geq 2$, and $q \geq 2$.*

PROOF. First, note that for $0 \leq k \leq G - 1$, $m_k = \frac{(2g-2)}{G}q - n'd'k \geq \frac{(2g-2)}{G}q - n'd'(G-1)$. So we need to show $\frac{(2g-2)}{G}q - n'd'(G-1) \geq n'd' - n' - d'$, which is equivalent to showing $(2g-2)q \geq nd - n - d$. Since $nd - n > nd - n - d$, by Lemma 2, we have $(2g-2)q \geq nd - d > nd - n - d$ for all (n, d, q) combinations except $(2, 5, 2)$ and $(2, 6, 2)$.

We compute the exceptional cases separately. If $(n, d, q) = (2, 5, 2)$, then $g = 2$ and $(2g-2)q = 4 \geq 3 = nd - n - d$. If $(n, d, q) = (2, 6, 2)$, then $g = 2$ and $(2g-2)q = 4 \geq 4 = nd - n - d$. Thus, the bound holds for the exceptional cases as well. □

For S_k, since we are considering $i \geq 0$ and $0 \leq j' < n'$, and since $m_k \geq n'd' - n' - d'$ for all k, our ordered pairs $(i, j' + kn')$ are in one-to-one correspondence with the (n', d')-representable numbers in the interval $[0, m_k]$. And since $m_k \geq n'd' - n' - d'$, all of the $(n'-1)(d'-1)/2$ (n', d')-non-representable numbers are in

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this interval as well. Thus S_k contains $|S_k| = m_k + 1 - (n'-1)(d'-1)/2$ ordered pairs.

Then

$$\begin{aligned} W_{1,k} &= \sum_{(i,j)\in S_k} (n'i + d'j) \\ &= \sum_{(i,j'+kn')\in S_k} (n'i + d'j' + n'd'k) \\ &= n'd'k \cdot |S_k| + \sum_{(i,j'+kn')\in S_k} (n'i + d'j')\,. \end{aligned}$$

The summation is the sum of the (n', d')-representable numbers from 0 to m_k. We calculate this by summing all of the integers from 0 to m_k and subtracting the (n', d')-non-representable integers, which all lie in this interval. Using Proposition 3, the summation is $m_k(m_k+1)/2 - (n'-1)(d'-1)(2n'd' - n' - d' - 1)/12$. Thus,

$$\begin{aligned} W_{1,k} = n'd'k\,&(m_k + 1 - (n'-1)(d'-1)/2) \\ &+ m_k(m_k+1)/2 - (n'-1)(d'-1)(2n'd' - n' - d' - 1)/12, \end{aligned}$$

so

$$\begin{aligned} W_1 =& G\sum_{k=0}^{G-1}\left[n'd'k\left(m_k + 1 - \frac{(n'-1)(d'-1)}{2}\right) + \frac{m_k(m_k+1)}{2}\right. \\ &\left. - \frac{(n'-1)(d'-1)(2n'd' - n' - d' - 1)}{12}\right]. \end{aligned}$$

To evaluate this sum, we need the following calculations which are straightforward to compute.

- $\sum_{k=0}^{G-1} m_k = (2g-2)q - \frac{n'd'G(G-1)}{2}$.
- $\sum_{k=0}^{G-1} m_k^2 = \frac{(2g-2)^2}{G}q^2 - (2g-2)(G-1)d'n'q + \frac{d'^2n'^2(G-1)G(2G-1)}{6}$
- $\sum_{k=0}^{G-1} km_k = (g-1)(G-1)q - \frac{d'n'(G-1)G(2G-1)}{6}$.

Simplifying the resulting expression, we find

$$W_1 = \frac{(2g-2)^2}{2}q^2 + (g-1)Gq + \frac{G^2 - 1 - (n-1)(d-1)(2nd - n - d - 1)}{12},$$

which completes the proof of Proposition 4. □

PROPOSITION 5. *Let $n, d, q \in \mathbb{N}$ such that $n < d$ and $q \geq 2$. Let*

$$D(a,b,c) = \sum_{j=0}^{c-1}\left\{\frac{a+bj}{c}\right\} j.$$

Then

$$W_2 = (d-1)\left((n-1)\left((g-1)q + \frac{-2nd + 3n + d}{6}\right) - D(-(d+G)q, -d, n)\right).$$

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PROOF. We will use Lemma 2, so we first assume $(n,d,q) \notin \{(2,5,2),(2,6,2)\}$. For $W_2 = (d-1)\sum_{(i,j)\in S} j$, we have

$$W_2 = (d-1)\sum_{j=0}^{n-1}\sum_{i=0}^{I_j} j,$$

for $I_j = \left\lfloor \frac{(2g-2)q-dj}{n} \right\rfloor$. By Lemma 2, $I_j \geq 0$ so $W_2 = (d-1)\sum_{j=0}^{n-1}(I_j+1)j$. Since $\lfloor x \rfloor = x - \{x\}$,

$$W_2 = (d-1)\sum_{j=0}^{n-1}\left(\frac{(2g-2)q-dj}{n} - \left\{\frac{(2g-2)q-dj}{n}\right\} + 1\right) j.$$

Note that $\left\{\frac{(2g-2)q-dj}{n}\right\} = \left\{\frac{(nd-n-d-G)q-dj}{n}\right\} = \left\{\frac{(-d-G)q-dj}{n}\right\}$.

Expanding out, we get

$$\begin{aligned} W_2 =& (g-1)(d-1)(n-1)q + \frac{(d-1)(n-1)}{6}(3n-d(2n-1)) \\ &- (d-1)\sum_{j=0}^{n-1}\left\{-\frac{(d+G)q+dj}{n}\right\} j, \end{aligned}$$

which can be rearranged to give the desired result.

Finally, if $(n,d,q) \in \{(2,5,2),(2,6,2)\}$, then $S = \{(0,0),(1,0),(2,0)\}$, and so $W_2 = \sum_{(i,j)\in S} j = 0$. We get the same value if we plug each these (n,d,q) triples into the above formula for W_2. □

REMARK. The summation $D(a,b,c)$ is related to a Dedekind sum. There is no closed form for such sums, though there is a reciprocity law. For a general reference, see [**9**].

Finally, we can combine and simplify $W_1 - W_2 - W_3$. Note that the q^2 and q terms (other than in the summation) cancel. With further manipulation, we have our main result.

THEOREM 2. *Let C be given in affine coordinates by $y^n = f(x)$ for $f(x)$ a separable polynomial of degree $d > n$. Let $G = \gcd(n,d)$, and let $q \in \mathbb{Z}$ with $q \geq 2$. For any root α of $f(x)$, let $B = (\alpha, 0)$ be a branch point.*

The q-weight of B is $w^{(q)}(B) =$

$$\begin{aligned} w^{(q)}(B) =& \frac{1}{24}\left((n-1)(d-1)(n+1)(d-7) + 12g(G+1) + 5(G^2-1)\right) \\ &+ (d-1)\cdot D(-(d+G)q, -d, n) \end{aligned}$$

Note that, for given values of n and d, the q-weight of B depends only on the value of q modulo n.

We will give results for some combinations of n and d in the corollaries below. First, we consider the case where $\gcd(n,d) = 1$.

COROLLARY 1. *If $\gcd(n,d) = 1$,*

$$w^{(q)}(B) = \frac{g}{12}(n+1)(d-7) + g + (d-1)\cdot D(-(d+1)q, -d, n).$$

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Fix n and d (with any gcd). If one varies q, then one sees the value of $w^{(q)}(B)$ depends only on the congruence class of q modulo n/G. Further, if $d \equiv -G \pmod{n}$, then the summation term simplifies to $\sum_{j=0}^{n-1} \left\{ \frac{Gj}{n} \right\} j$, for which there is a closed form.

COROLLARY 2. *If $d \equiv -G \pmod{n}$, then $w^{(q)}(B)$ doesn't depend on q. In particular,*

$$w^{(q)}(B) = \frac{1}{24}\Big((n-1)(d-1)(n+1)(d-7) + 12g(G+1) + 5(G^2-1) + 2(d-1)(n-G)(3n+n'-2)\Big).$$

PROOF. The summation term is $\sum_{j=0}^{n-1} \left\{ \frac{Gj}{n} \right\} j, = \sum_{j=0}^{n-1} \left\{ \frac{j}{n'} \right\} j$. Each j can be written uniquely as $j = j' + kn'$ for $0 \le k < G$ and $0 \le j' < n'$. Thus, the summation is $\sum_{k=0}^{G-1} \sum_{j'=0}^{n'-1} \left\{ \frac{j'}{n'} \right\} j = \sum_{k=0}^{G-1} \sum_{j'=0}^{n'-1} \left(\frac{j'^2}{n'} + j'k \right)$, which simplifies to $(n-G)(3n+n'-2)/12$. □

Combining the two corollaries above, we obtain the following.

COROLLARY 3. *If $d \equiv -1 \pmod{n}$, then*

$$w^{(q)}(B) = \frac{g(n+1)(d+1)}{12} = \frac{(n^2-1)(d^2-1)}{24}$$

for all $q \ge 2$.

COROLLARY 4. *If $n \mid d$, then*

$$w^{(q)}(B) = \frac{(n^2-1)(d^2-2d)}{24}.$$

PROOF. If $n \mid d$, then $G = n$ and $n \mid ((d+G)q + dj)$ for all j, so the summation is zero. Since $2g - 2 = nd - n - d - n$, we have $g = \frac{(d-2)(n-1)}{2}$. Plugging in, the result follows. □

4.2. Weights of points at infinity. If $n \mid d$, then there are n points at infinity in the smooth model of C, so these points are not branch points. However, we can still investigate their q-weights. If $\gcd(n,d) > 1$, then we need to know more about $f(x)$ to determine $w^{(q)}(P_m^\infty)$. We give a few examples to illustrate this.

In [**3**], the authors consider curves of the form $y^2 = f(x) = x^6 + ax^4 + bx^2 + 1$, where a, b are parameters and $f(x)$ is separable. In the non-singular models of these curves, there are $G = \gcd(n,d) = 2$ points at infinity P_1^∞ and P_2^∞. If $4b = a^2$, then $w^{(3)}(P_1^\infty) = w^{(3)}(P_2^\infty) = 2$. If $4b \ne a^2$, then $w^{(3)}(P_1^\infty) = w^{(3)}(P_2^\infty) = 0$.

In [**13**, Lemma 4 and Proposition 3], the authors consider hyperelliptic curves of genus 3 of the form $y^2 = f(x)$ where $\deg(f) = 8$. In the non-singular models of these curves, there are $G = \gcd(n,d) = 2$ points at infinity P_1^∞ and P_2^∞. If C is given by $y^2 = x^8 + x^6 + 16x^4 + x^2 + 1$, then $w^{(2)}(P_1^\infty) = w^{(2)}(P_2^\infty) = 1$. If C is given by $y^2 = x^8 + x^4 + 1$, then $w^{(2)}(P_1^\infty) = w^{(2)}(P_2^\infty) = 3$.

Thus, simply knowing n and d is not enough to calculate the q-weight of the points at infinity. However, there are some cases where we can get a result.

First, if $d = n+1$, then the lone point at infinity is a nonsingular branch point, so it will have the same q-weight as the affine branch points. By Corollary 3, since

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$d \equiv -1 \pmod{n}$, $w^{(q)}(B) = \frac{(n^2-1)(d^2-1)}{24}$ for $q \geq 2$, so we will have $w^{(q)}(P^\infty) = \frac{(n^2-1)(d^2-1)}{24}$ for $q \geq 2$ as well. This is a special case of the more general result when $\gcd(n,d) = 1$.

THEOREM 3. *Suppose C is a curve of genus $g \geq 2$ given by the affine equation $y^n = f(x)$ for $f(x)$ a separable polynomial of degree d where $n < d$ and $\gcd(n,d) = 1$. Let P_1^∞ be the lone point at infinity in the non-singular model of C. Then*

$$w^{(q)}(P_1^\infty) = \begin{cases} \dfrac{g(n+1)(d+1)}{12} - g = \dfrac{(n^2-1)(d^2-1)}{24} - g & \text{if } q = 1, \\ \dfrac{g(n+1)(d+1)}{12} = \dfrac{(n^2-1)(d^2-1)}{24} & \text{if } q \geq 2. \end{cases}$$

PROOF. For $q = 1$, the formula is given at the end of the proof of [**16**, Theorem 8].

For $q \geq 2$ and $G = 1$, let $\mathfrak{B}_{n,d,q}$ be as in Section 3, and again let $S = \{(i,j) \in \mathbb{Z}^2 : i \geq 0, 0 \leq j < n, ni + dj \leq (2g-2)q\}$. Then $f_{i,j} \in \mathfrak{B}_{n,d,q}$ if and only if $(i,j) \in S$. Recall that $\operatorname{ord}_{P_m^\infty}(f_{i,j}) = (2g-2)q - (ni + dj)$. These orders of vanishing are unique, so

$$w^{(q)}(P_m^\infty) = \left(\sum_{(i,j)\in S} \operatorname{ord}_{P_m^\infty}(f_{i,j}) \right) - \sum_{k=0}^{d_q - 1} k.$$

Since $|S| = d_q$,

$$w^{(q)}(P_m^\infty) = d_q(2g-2)q - \left(\sum_{(i,j)\in S} (ni + dj) \right) - \frac{(d_q - 1)d_q}{2}.$$

The summation, which we called W_1 in Equation 1, is evaluated in Proposition 4. Plugging this and d_q in, the expression simplifies to $w^{(q)}(P_m^\infty) = \frac{(n^2-1)(d^2-1)}{24}$. □

4.3. Branch weight. In the case where $\gcd(n,d) = 1$, we can calculate the total q-weight of the branch points (both affine and at infinity) for $q \geq 2$, which we denote BW_q.

COROLLARY 5. *Suppose $\gcd(n,d) = 1$, so $g = \frac{(n-1)(d-1)}{2}$. Then the total branch q-weight is given by $BW_q = d \cdot w^{(q)}(B) + w^{(q)}(P_1^\infty) =$*

$$d\left(\frac{g}{12}(n+1)(d-7) + g + (d-1) \cdot D(-(d+1)q, -d, n)\right) + g\frac{(n+1)(d+1)}{12}.$$

Rewritten in terms of g, we get

$$BW_q = \frac{n+1}{3(n-1)^2}\left(g^3 - 2g^2(n-1) - g(n-1)^2\right) + d(d-1) \cdot D(-(d+1)q, -d, n).$$

From Proposition 1, we know the total weight of the q-Weierstrass points, for $q \geq 2$, is $g(g-1)^2(2q-1)^2$. We can now calculate the proportion of q-weight of the branch points.

PROPOSITION 6. *Fix n and let $q \geq 2$. Then*

$$\liminf_{d\to\infty} \frac{BW_q}{g(g-1)^2(2q-1)^2} \geq \frac{n+1}{3(n-1)^2(2q-1)^2}.$$

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If we restrict to values of d that are relatively prime to n then

$$\lim_{d\to\infty,(n,d)=1} \frac{BW_q}{g(g-1)^2(2q-1)^2} = \frac{n+1}{3(n-1)^2(2q-1)^2}.$$

PROOF. For general n and d, since we do not have an exact formula for the q-weight of the points at infinity, we can only say $BW_q \geq d \cdot w^{(q)}(B)$. Using the result from Theorem 2, since

$$\sum_{j=0}^{n-1} \left\{ -\frac{(d+G)q+dj}{n} \right\} j \leq \sum_{j=0}^{n-1} \frac{n-1}{n} j = \frac{(n-1)^2}{2},$$

in terms of d, the dominant term of $d \cdot w^{(q)}(B)$ is $d^3 \frac{(n-1)(n+1)}{24}$. Since g is on the order of $d(n-1)/2$, the dominant term of the denominator is $d^3 \frac{(n-1)^3(2q-1)^2}{8}$. The result follows.

For $\gcd(n,d)=1$, the lone point at infinity has weight $\frac{(d^2-1)(n^2-1)}{24}$. Thus, the dominant term of BW_q is precisely $d^3 \frac{(n-1)(n+1)}{24}$, and we thus have an equality if we take a limit involving integers d such that $\gcd(n,d)=1$. □

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